Classical Field Theory — Exercise Sheet 14

First-order systems and real structures

Hints and answers

All spinor entries commute. Use Lecture 14's Hermitian inner product, unitary Fourier convention and real form $\Omega(h,g) = 2\,\mathrm{Im}\int h^\dagger g$. Problems 1 and 3 supply new four-component matrices; Problem 4 returns to the actual two-component matrices of Lecture 14.

Exercise 14.1 (Four components in three space dimensions). Let I_2 be the two-by-two identity and put

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \Gamma^0 = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix}, \quad \Gamma^i = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix}.$$

Define $\beta_D = \Gamma^0$, $\alpha_D^i = \Gamma^0\Gamma^i$ and $H_D(k) = \alpha_D^i k_i + m\beta_D$, where $m \geq 0$.

(a) Verify every Clifford anticommutator for signature $(+,-,-,-)$. Prove that $H_D(k)$ is Hermitian and find its eigenvalues, multiplicities and orthogonal spectral projectors, including the exceptional $m = k = 0$ case.

(b) For $\mathrm{i}\psi_t = (-\mathrm{i}\alpha_D^i\partial_i + m\beta_D)\psi$, derive the conserved current $\rho = \psi^\dagger\psi$, $j^i = \psi^\dagger\alpha_D^i\psi$ and prove $|n\cdot j| \leq \rho$ for every unit vector n. Explain why verifying these matrices alone does not prove the spin representation or Lorentz-covariance theorem cited in Lecture 14.

Hint. First multiply the Pauli matrices. Square H_D before finding its projectors, and use the trace to determine their ranks.

Answer. For $\nu = \sqrt{|k|^2 + m^2} > 0$, the eigenvalues are $\pm\nu$, each twice, with projectors $(I_4 \pm H_D/\nu)/2$. At $m = k = 0$, $H_D = 0$. The current obeys $\partial_t\rho + \partial_i j^i = 0$ and $|n\cdot j| \leq \rho$.

Exercise 14.2 (A constant symmetric-hyperbolic system). Let $A^1, \dots, A^d$ be constant Hermitian N-by-N matrices and B a constant skew-Hermitian matrix. For complex columns consider

$$w_t + A^i\partial_i w + Bw = 0, \qquad c_* = \sup_{|n|=1} \|n_i A^i\|_{\mathrm{op}}.$$

Here the operator norm is the Euclidean matrix norm.

(a) Construct the unique distributional solution in $C(\mathbb{R}; L^2)$ for each L^2 datum and prove norm preservation. Derive the smooth local current.

(b) Using Lecture 5's moving-ball argument, prove that data supported in a closed set K remain supported in $\{x : \mathrm{dist}(x,K) \leq c_*|t|\}$. Give the limiting argument for rough data and include $c_* = 0$.

(c) For $d = N = 2$, take $A^1 = \sigma_1$, $A^2 = 2\sigma_3$ and $B = \mathrm{i}\sigma_2$. Compute c_* and exhibit a column and normal attaining the current bound. Would Hermitian $B = I_2$ preserve the same norm?

Hint. The Fourier generator $\mathrm{i}k_iA^i + B$ is skew-Hermitian. On a shrinking ball use $j\cdot n \geq -c_*\rho$ at the boundary. For rough data pass local squared norms, rather than boundary traces, to the limit.

Answer.

$$\widehat{w}(t,k) = e^{-t(\mathrm{i}k_i A^i + B)}\widehat{w}(0,k), \qquad \rho = w^\dagger w, \quad j^i = w^\dagger A^i w.$$

The example has $c_* = 2$, attained by $w = e_1$, $n = e_2$. For $B = I_2$ the total squared norm decays as e^{-2t}.

Exercise 14.3 (Weyl sectors and a mass coupling). Use the four-component matrices of Problem 1. Define $\Gamma^5 = \mathrm{i}\Gamma^0\Gamma^1\Gamma^2\Gamma^3$ and $P_{\mathrm{ch},\pm} = (I_4 \pm \Gamma^5)/2$. Compute Γ^5 and its two eigenspaces. Write the Dirac Hamiltonian in the coordinates

$$\psi = \frac{1}{\sqrt{2}}\begin{pmatrix} u+v \\ u-v \end{pmatrix}, \qquad u, v \in \mathbb{C}^2.$$

Determine when the two chiral subspaces are invariant under evolution, and derive the two Weyl equations in the massless case.

Hint. Compare $[\Gamma^5, \alpha_D^i]$ with $\Gamma^5\beta_D + \beta_D\Gamma^5$. The displayed change of coordinates is unitary.

Answer.

$$\Gamma^5 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}, \qquad H_{\mathrm{ch}}(k) = \begin{pmatrix} \sigma\cdot k & mI_2 \\ mI_2 & -\sigma\cdot k \end{pmatrix}.$$

The separate chiral subspaces are invariant exactly when $m = 0$.

Exercise 14.4 (A commuting Majorana-type real subspace). Return to Lecture 14's two-component matrices $\alpha = -\sigma_3$, $\beta = \sigma_1$ and define the real-linear, antilinear-in-complex-scalars map

$$C\psi = \sigma_3\overline{\psi}.$$

Prove $C^2 = I$, characterize its fixed space in L^2, and show that Dirac evolution preserves it. Derive the evolution of its two real entries. Restrict the real form Ω to that fixed space. Is the result nondegenerate? Explain why this calculation does not give a classical commuting replacement for a quantum Majorana field or its statistics.

Hint. The matrix σ_3 commutes with α and anticommutes with β. Test the imaginary part of the inner product of two fixed columns.

Answer. Fixed columns have the form $\psi = (a, \mathrm{i}b)^T$ with a, b real, and obey $a_t = a_x + mb$, $b_t = -b_x - ma$. The restricted form is identically zero.